minimize shipping cost

maximize revenue

Cost v. Price Interpretation

Primal (transport plan):

- Sender wants to **minimize shipping cost**
- π_{ij} = how much to ship from x_i to y_j
- c_{ij} = standard shipping cost to ship one unit from x_i to y_j
- Objective = min total shipping cost: $\min_{\pi \geq 0} \sum_{i,j} c_{ij} \pi_{ij}$
- Constraint = supply/demands are met

Dual (potentials):

- Vendor wants to **maximize revenue**
- $\mathbf{f}_i$ = price set by vendor to collect 1 unit from x_i
- $\mathbf{g}_i$ = price set by vendor to deliver 1 unit to y_j
- Objective = maximize total revenue: $\max_{\mathbf{f},\mathbf{g}} \sum_i \mathbf{f}_i \mathbf{a}_i + \sum_j \mathbf{g}_j \mathbf{b}_j$
- Constraint = $\mathbf{f}_i + \mathbf{g}_j \leq c_{ij} \implies$ price never exceeds shipping cost

Dual Problem - General Case

Primal Problem

$$\inf_{\pi \in \Pi(\alpha,\beta)} \int_{X \times Y} c(x, y)\, \mathrm{d}\pi(x, y) \qquad \text{where}$$

$$\Pi(\alpha, \beta) = \{\pi \in \mathscr{P}(\mathscr{X} \times \mathscr{Y}) \mid P_{\mathscr{X}\sharp}\pi = \alpha, P_{\mathscr{Y}\sharp}\pi = \beta\}$$

Dual Problem